

Cancer immunotherapy using dendritic cell-derived exosomes.

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Dendritic cells (DCs) are the most potent antigen presenting cells and the only ones capable of inducing primary cytotoxic immune responses. We found that DCs secrete a population of membrane vesicles, called exosomes. Exosomes are 60-80 nm vesicles of endocytic origin. The protein composition of exosomes was subjected to a systematic proteomic analysis. Besides MHC and co-stimulatory molecules, exosomes bear several adhesion proteins, most likely involved in their specific subjected to targeting. We also found that exosomes accumulate several cytosolic factors, probably involved in their endosomal biogenesis. Like DCs, exosomes induced immune responses in vivo. Indeed, a single injection of DC-derived exosomes sensitized with tumor peptides induced potent anti tumor immune responses in mice and the eradication of established tumors. Tumor-specific cytotoxic T lymphocytes were found in the spleen of exosome-treated mice, and the anti tumor effect of exosomes was sensitive to in vivo depletion of CD8+ T cells. These results show that exosomes induce potent anti tumor effects in vivo, and strongly support the implementation of human DC-derived exosomes for cancer immunotherapy.

Antigen presentation in EAE: role of microglia, macrophages and dendritic cells.

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Experimental autoimmune encephalomyelitis (EAE), a well-established model of multiple sclerosis, is characterised by microglial activation and lymphocytic infiltration. Lymphocytic activation through the antigen presentation process involves three main signals, the first provided by the engagement of major histocompatibility complex molecules (MHC) with the receptor of T-cells (TCR), the second by the binding of co-stimulatory molecules and the third by the secretion or expression of T-cell polarising molecules in specific populations of antigen presenting cells (APC). Microglial cells are considered to be the main APC population in the central nervous system (CNS). Specifically in EAE an increase in MHCs, co-stimulatory molecules and different T-cell polarising factors have been reported in microglia. However, a growing number of evidences suggest that dendritic cells (DCs), the main APC in the peripheral immune system, may also participate in the regulation of T-cell responses within the CNS. In this review we summarize the principal knowledge regarding microglial/macrophage function in EAE and their role in T-cell modulation, as well as the participation of DCs in the immune response associated to this disease.

Dendritic Cell Differentiation from Human Induced Pluripotent Stem Cells: Challenges and Progress.

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Dendritic cells (DCs) are the major antigen-presenting cells of the immune system responsible for initiating and coordinating immune responses. These abilities provide potential for several clinical applications, such as the development of immunogenic vaccines. However, difficulty in obtaining DCs from conventional sources, such as bone marrow, peripheral blood, and cord blood, significantly hinders routine application. The use of human induced pluripotent stem cells (hiPSCs) is a valuable alternative for generating sufficient numbers of DCs to be used in basic and preclinical studies. Despite the many challenges that must be overcome to achieve an efficient protocol for obtaining the major DC types from

hiPSCs, recent progress has been made. In this study, we review the current state of developing DCs from hiPSCs, as well as the key elements required to enable the routine use of hiPSC-derived DCs in preclinical and clinical assays.

The Role of Dendritic Cells in TB and HIV Infection.

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Dendritic cells are the principal antigen-presenting cells (APCs) in the host defense mechanism. An altered dendritic cell response increases the risk of susceptibility of infections, such as *Mycobacterium tuberculosis* (M. tb), and the survival of the human immunodeficiency virus (HIV). The altered response of dendritic cells leads to decreased activity of T-helper-1 (Th1), Th2, Regulatory T cells (Tregs), and Th17 cells in tuberculosis (TB) infections due to a diminishment of cytokine release from these APCs, while HIV infection leads to DC maturation, allowing DCs to migrate to lymph nodes and the sub-mucosa where they then transfer HIV to CD4 T cells, although there is controversy around this topic. Increases in the levels of the antioxidant glutathione (GSH) plays a critical role in maintaining dendritic cell redox homeostasis, leading to an adequate immune response with sufficient cytokine release and a subsequent robust immune response. Thus, an understanding of the intricate pathways involved in the dendritic cell response are needed to prevent co-infections and co-morbidities in individuals with TB and HIV.

Dendritic cells.

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Dendritic cells (DC) are the most effective or 'professional' of the antigen-presenting cells (APC) that initiate primary immune responses. They are located at surveillance sites where they capture and process antigens. They then initiate and regulate T- and B-cell responses by expressing lymphocyte costimulatory molecules, migrating to lymphoid organs and secreting biologically active molecules. Dendritic cells not only activate lymphocytes to induce the immune response, but they also minimize autoimmune reactions by tolerizing T cells to self-antigens. Recent Phase I and II clinical studies have shown promise in the use of antigen-pulsed autologous DC for vaccination of cancer patients. Dendritic cells also have applications in preventing rejection after transplantation, immunization against viral infections and immunosuppression in autoimmune diseases.

Cytokine-overexpressing dendritic cells for cancer immunotherapy.

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Dendritic cells (DCs), the main type of antigen-presenting cells in the body, act as key mediators of adaptive immunity by sampling antigens from diseased cells for the subsequent priming of antigen-specific T and B cells. While DCs can secrete a diverse array of cytokines that profoundly shape the immune milieu, exogenous cytokines are often needed to maintain the survival, proliferation, and differentiation of DCs, T cells, and B cells. However, conventional cytokine therapies for cancer treatment are limited by their low therapeutic benefit and severe side effects. The overexpression of cytokines in

DCs, followed by paracrine release or membrane display, has emerged as a viable approach for controlling the exposure of cytokines to interacting DCs and T/B cells. This approach can potentially reduce the necessary dose of cytokines and associated side effects to achieve comparable or enhanced antitumor efficacy. Various strategies have been developed to enable the overexpression or chemical conjugation of cytokines on DCs for the subsequent modulation of DC-T/B-cell interactions. This review provides a brief overview of strategies that enable the overexpression of cytokines in or on DCs via genetic engineering or chemical modification methods and discusses the promise of cytokine-overexpressing DCs for the development of new-generation cancer immunotherapy.

Anti-leucocyte common (CD45) antibodies inhibit dendritic cell stimulation of CD4 and CD8 T-lymphocyte proliferation.

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Dendritic cells (DC) act as primary antigen-presenting cells and probably initiate T-cell responses via the formation of DC-T-cell clusters. Both DC and T cells express members of the leucocyte common (CD45) family of molecules as major cell membrane proteins. The possibility that the leucocyte common antigens (LCA) play a functional role in the cellular events of a primary T-cell response was tested, by examining the effect of CD45 antibodies on the T-cell proliferative response induced by stimulating with allogeneic DC in the mixed leucocyte reaction (MLR). The presence of individual CD45 monoclonal antibodies during the MLR inhibited T-cell proliferation only weakly, but a mixture of monoclonal antibodies against different epitopes of the LCA had a more profound inhibitory effect. The effect of the CD45 antibodies was maximal during the first 24 hr and preincubation experiments indicated that the major blocking effect occurred at the T-cell level, although some inhibition after preincubation with DC was noted. Purified allogeneic CD8 T-cell responses were more susceptible to inhibition than purified CD4 cell responses, but no difference in the inhibition of the CD45RA (CMRF-11) or CD45RO (UCHL1) LCA-positive T-cell subsets was observed. The presence of CD45 antibodies did not inhibit DC-T-cell clustering but reduced cluster stability, suggesting that certain epitopes of the LCA may be involved in the process of cell adhesion.

Expression of relB is required for the development of thymic medulla and dendritic cells.

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Dendritic cells (DC) derived from bone marrow are critical in the function of the immune system, for they are the primary antigen-presenting cells in the activation of T-lymphocyte response. Their differentiation from precursor cells has not been defined at a molecular level, but recent studies have shown an association between expression of the relB subunit of the NF-kappa B complex and the presence of DC in specific regions of normal unstimulated lymphoid tissues. Here we show that relB expression also correlates with differentiation of DC in autoimmune infiltrates in situ, and that a mutation disrupting the relB gene results in mice with impaired antigen-presenting cell function, and a syndrome of excess production of granulocytes and macrophages. Thymic UEA-1+ medullary epithelial cells from normal mice show striking similarities to DC and, interestingly, these cells are also absent in relB mutant mice. Taken together, these results suggest that relB is critical in the coordinated activation of genes necessary

for the differentiation of two unrelated but phenotypically similar cells (DC and thymic UEA-1+ medullary epithelial cells) and is therefore a candidate for a gene determining lineage commitment in the immune system.

Dendritic cell-based vaccines and therapies for cancer.

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Dendritic cells (DCs) are the most potent professional antigen-presenting cells (APCs), and are capable of stimulating naive T cells and driving primary immune responses. Due to this unique property, DCs represent both vectors and targets for immunological intervention in numerous diseases and are optimal candidates for vaccination protocols in cancer. This review provides an overview of DC immunobiology and the integration of these important APCs in immunotherapies for cancer.

Dendritic Cell Generation from Highly Purified CD14+Monocytes.

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Dendritic cells (DC) play a pivotal role in the function of the immune system, for they are the primary antigen-presenting cells (APC) in the activation of naive T-lymphocyte responses (1). Recent studies have uncovered complexity in the DC lineage with several subsets, functions, and maturational stages. Although it is generally accepted that human DC derive from hematopoietic progenitor cells (2-9), it is not clear at present whether DC cells and their precursors represent a separate hematopoietic lineage or whether DC should be seen as specialized macrophages with particular morphological, molecular, and functional features. Several lines of evidence point to DC and monocytes/ macrophages being offspring of the same CD34(+) hematopoietic progenitor cell (3-5,12-14, and reviewed in ([10,11].) DC committed precursor cells have also been identified in peripheral blood (15-18).
